

FlyRank AI Internship — Capstone Report

General AI Fluency · Impact Project (FL)

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Internship Track: General AI Fluency

Week: 6

Workload: 12 hours

Executive Summary

For my Week 6 capstone project, I built and deployed a Postpartum AI Companion — a warm, empathetic 24/7 AI health assistant supporting new mothers through postpartum recovery. The project encompasses all three capstone requirements: mastering the AI stack, building a personal brand website, and shipping a functional personal agent.

Live URLs:

- GitHub: <https://github.com/JIYA-YDV/flyrank-internship-tracker/tree/main/Capstone/General%20AI%20Fluency%20Impact%20Project>

- Chatbot: https://huggingface.co/spaces/YDVJIYA/postpartum_AI_companion

1. Problem Statement

New mothers globally face a critical support gap:

- Healthcare systems focus almost entirely on the baby
- Postpartum depression affects 1 in 7 women
- Most support resources are only available 9-5
- Cultural stigma prevents open conversation
- 3 AM is the loneliest time — and no one is awake

My hypothesis: An empathetic, safety-conscious AI companion — available 24/7, judgment-free, always pointing toward professional care — could bridge this gap meaningfully.

2. Solution Overview

A conversational AI agent named "Jiya" specializing in:

- Postpartum physical recovery
- Mental health (baby blues, PPD, anxiety, psychosis awareness)
- Breastfeeding support
- Newborn care basics
- Crisis detection with hotline referral

Key differentiator: Empathy-first responses that validate feelings BEFORE providing information — designed to feel like a knowledgeable friend, not a clinical textbook.

3. Technical Architecture

High-Level Flow

User (Web Browser)

↓

Portfolio Website (GitHub Pages)

↓ iframe embed

Gradio Chat UI (Hugging Face Spaces)

↓

Python Application with Safety Prompts

↓

Groq API → Llama 3.3 70B

↓

Streaming Response

Tech Stack

Layer	Technology	Justification
LLM	Llama 3.3 70B (Groq)	Free, fast (<1s), quality on par with GPT-4
Framework	Gradio 5.x	Native HF Spaces support, easy custom theming
Language	Python 3.11	Industry standard for AI workflows
Hosting (Agent)	Hugging Face Spaces (ZeroGPU)	Free public deployment
Hosting (Site)	GitHub Pages	Free static hosting
Version Control	Git + GitHub	Portfolio-visible source code

4. Development Process

Timeline (12 hours)

- Hours 1-2: Project setup, AI stack documentation
- Hours 3-5: Agent development, prompt engineering
- Hours 6-8: Portfolio website design and build
- Hours 9-11: Deployment and integration
- Hour 12: Documentation and polish

Real-World Pivots

Original Plan	What Changed	Reason
OpenAI GPT-4o	Groq Llama 3.3 70B	Credits exhausted mid-build
Chainlit UI	Gradio	Chainlit lacks free deployment
Docker on HF	Gradio SDK	Docker requires paid plan

5. Safety Design

Given the sensitive medical context, I implemented multiple safety layers:

1. System Prompt Rules:

- Never diagnose
- Never advise stopping medication
- Always end responses with a doctor consultation reminder

2. Crisis Detection:

- Auto-response for self-harm mentions
- Provides Maternal Mental Health Hotline: 1-833-943-5746
- Directs to emergency services when appropriate

3. UI Disclaimers:

- Prominent medical disclaimer on every page
- Emergency contact info always visible

6. Key Learnings

Technical

- API migration is a real skill — Switching LLM providers under time pressure
- File encoding matters — UTF-8 vs Windows-1252 broke deployment for hours
- Free tier constraints breed creativity — Forced me to learn HF Spaces deeply
- Streaming responses > blocking — Perceived latency drops dramatically

Product

- Empathy is design work — The system prompt was harder to write than the code
- Safety must be architected in — Not bolted on
- Portfolio > Project — One live deployed thing beats ten local demos

Personal

- Git commits tell your story — 15+ conventional commits show my process
- Pivoting isn't failure — It's the actual skill being tested

7. Deliverables Summary

Requirement	Deliverable	Location
AI Stack Mastery	Detailed markdown doc	/ai-stack.md
Personal Brand Website	Live portfolio hub	GitHub Pages URL above
Personal Agent	Deployed chatbot	Hugging Face URL above
Documentation	README + this report	/README.md,/docs/capstone-report.pdf
Version Control	15+ meaningful commits	GitHub repository

8. Future Roadmap

- v2 Features:
 - RAG with vetted medical sources (WHO, ACOG)
 - Multi-language support (Hindi priority)
 - Voice I/O for hands-free use
 - Symptom tracking with weekly insights
 - Partner-support mode
- Long-term Vision:
 - Partner with maternal health orgs
 - Fine-tune model on postpartum-specific data
 - Add anonymous usage analytics

9. Reflection

Six weeks ago, "AI" felt like magic. Now it feels like engineering with constraints, ethics, and empathy.

I learned that:

- The best AI projects solve real problems for real people
- Real deployment is where 80% of the learning happens
- Failure logs are the best teachers

- Empathy is a technical skill

This capstone gave me confidence that I can build meaningful AI products — from prompt to deployment — under real-world constraints.

10. Acknowledgments

- FlyRank — for the internship framework and capstone challenge
- Groq — for free, blazing-fast LLM inference
- Hugging Face — for democratizing AI deployment
- Claude (Anthropic) — as my pair-programming companion during this build

Signed,

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